Is YouTube Feminist or Anti-Feminist? A Comparative Sock-Puppet Experiment on South Korean YouTube Recommendations

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Abstract

South Korean radical feminist groups such as Megalia and anti-feminist groups like New Men's Solidarity have turned YouTube and other social media platforms into a battleground for polarized identity formation. This study examines how YouTube's recommendation system structures exposure to feminist and anti-feminist content in South Korea, focusing on ideological and topical bias across user conditions. We employ a sock-puppet audit design with three simulated user profiles (neutral, feminist, and anti-feminist). For each condition, we collect the top 20 recommended videos for 20 seed videos. A total of 615 videos are coded into ideological and topical categories using LLM-assisted classification. The results show evidence of both ideological and topical bias in recommendations. Anonymous users are exposed to more anti-feminist content, while feminist-conditioned accounts exhibit stronger ideological reinforcement than anti-feminist accounts. We also find substantial differences in topical distribution across conditions, suggesting that recommendation algorithms shape not only ideological exposure but also issue emphasis, potentially reinforcing more emotional and conflict-oriented content environments.

Keywords: recommendation algorithm, echo chambers, gender conflict, YouTube, South Korea

1 Introduction

The concepts of filter bubbles and echo chambers have been central to debates about algorithmic personalization, polarization, and the fragmentation of public discourse on platforms like YouTube. Filter bubbles describe the personalization effect of algorithmic curation that limits exposure to diverse viewpoints[1], while the related concept of the echo chamber illustrates an ideologically homogenious social sphere where individuals engage primarily with like-minded others [2]. Earlier research posited that recommendation algorithms reinforce ideological segregation by exposing users primarily to attitude-consistent content [3–6], but recent findings have questioned and partially refuted the universality and magnitude of these effects [7–17]. Despite substantial disagreement in the literature about the existence and the polarizing effect of online echo chambers, scholars in the field increasingly acknowledge that content exposure on social media platforms is not solely depending on algorithmic curation, but strongly depends on user behaviour [11, 13, 18, 19].

To account for this complexity, Geschke et al. [20] as well as He and Fan [19] propose to understand selective exposure as a three dimensional phenomenon, shaped by the interplay of user agency, social networks, and algorithms. Accordingly, echo chambers can form organically through cognitive and social dynamics, but algorithms can amplify these tendencies by impacting not only content recommendations but also users' media exposure [9, 10]. Furthermore, Ahmmad et al. [21] point out that social media platforms like YouTube are not merely personalized media environments, but also sites of identity formation and cultural belonging. In this sense, algorithmic curation can contribute to the development of shared vocabularies, humor, and symbolic boundaries that reinforce in-group identities. Consequently, studying algorithmic content curation remains particularly relevant in the context of polarized and identity-driven online communities, especially among younger users for whom digital media environments are closely tied to processes of social belonging.

Furthermore, comparative studies suggest that recommendation systems push users differently across ideological spectra—however scholars have presented contradictory findings. Ibrahim et al. find that YouTube's algorithm in the US pulls users away from far-right content more strongly than from far-left content and that YouTube recommends left-leaning content by default [22]. On the other hand, Haroon et al. find evidence for a stronger filter bubble effect for far-right users [23]. Focusing on South Korean news media on YouTube, Park and Park [24] find that selective exposure is stronger among consumers of conservative content than progressive content. Specifically, large differences occur in how much contrasting arguments are recommended for both groups: only 7 % of video recommendations for conservative seed videos contained contrasting arguments, compared to approximately 25 % for progressive seed videos. Park and Park [24] conclude that their research "provides empirical evidence of YouTube's efforts to promote authoritative voices" (p. 207). The presented mixed empirical evidence underscores the need for further research to determine whether YouTube's algorithm exhibits directional bias depending on different ideological algorithmic personalization.

In light of the above, South Korea's gender conflict emerges as a highly relevant case study. Young Koreans are strongly divided along the lines of attitudes towards gender equality [25]. While young women strongly support and demand gender equality measures, many young men in South Korea increasingly reject feminism, sparking an "unprecedented gender war" that predominantly plays out online [25, 26]. Feminist and anti-feminist communities on YouTube use the platform not only to consume content but to construct and reinforce collective identities. For instance, the online feminist group Megalia gained widespread popularity for its provocative "mirroring" strategy, turning misogynistic language against men to combat misogyny [26]. In contrast, male followers of anti-feminist organizations like New Men's Solidarity view themselves as victims of gender discrimination and often blame feminists for their grievances, sometimes resulting in violent attacks against women [27, 28]. Thus, South Korea's gender conflict represents a highly polarized, and identity-driven context—ideal for examining potentially asymmetric algorithmic curation on YouTube, particularly among younger users for whom digital platforms are central for their identity formation.

Against this backdrop, we ask: *Do YouTube video recommendations for feminist and anti-feminist profiles in South Korea differ in their ideological and topical directional bias?* Based on the findings by Park & Park [24], we hypothesize that YouTube recommendations will display a stronger filter bubble effect within anti-feminist recommendation environments compared to feminist ones. We analyze not only ideological but also topical directional bias, because we want to capture the dynamics of gender conflict in South Korea in the specific language and societal context. Recent literature also argues that YouTube's recommendation algorithm directs users towards popular, entertaining, and less sensitive topics [29, 30]. To answer the research question, our study adopts a comparative sock-puppet audit methodology. Two YouTube accounts were systematically trained through repeated exposure to either feminist or anti-feminist Korean-language content, alongside a baseline incognito condition approximating minimal personalization. Recommendation outputs from ideologically distinct seed videos were subsequently analyzed according to ideological orientation and topical framing using a human–machine annotation approach combining manual coding procedures with GPT-4o-mini-assisted content classification. The sock-puppet approach is particularly suitable for examining algorithmic personalization because it allows for controlled and reproducible manipulation of browsing histories[9, 23]. Although such experimental profiles cannot capture the complexity of real-world media consumption, they provide a systematic method for identifying asymmetries in recommendation systems and assessing how algorithmic curation may supply material that contributes to ideological polarization.

The paper proceeds as follows. We first present our method and describe the data collection process. Second, we outline and analyze the results of our study. Lastly, we summarize and discuss our findings as well as their implications and limitations.

2 Method

To investigate how recommendation systems respond to different ideological viewing patterns, we conducted a sock puppet experiment under three recommendation conditions: a baseline incognito condition, a feminist-profile condition, and an anti-feminist-profile condition. Two separately created YouTube accounts were systematically trained through repeated exposure to manually labeled ideologically aligned content (see Supplementary Material A for the full training log). Recommendation outputs were then collected from a set of seed videos under each condition. To analyze recommendation characteristics, GPT-4o-mini was used to classify the ideological orientation and topical framing of recommended videos based on a manually developed codebook.

2.1 Topic Context and Seed Video Selection

Gender discourse in South Korea is highly dynamic and context-dependent. Discussions frequently extend beyond explicitly ideological content and are embedded within broader social debates. In addition, online gender conflicts often rely on culturally specific symbols, slang, and coded expressions that carry implicit ideological meanings within Korean digital culture [31]. To capture these diverse dynamics, we first compiled a list of Korean-language keywords commonly associated with gender-related debates and online gender conflict in South Korea (see Supplementary Material B for the full keyword list and translation rationale).

Using these keywords, we conducted an exploratory mapping of gender-related discourse on YouTube through Korean-language keyword searches (see Supplementary Material C for the full video selection process). This process identified five recurring thematic domains that consistently appeared across both feminist and anti-feminist content ecosystems. These domains are used in our later video categorization.

Domain	Keywords	Associated Conflict
Sexual Violence	molka, Nth Room, deepfake pornography, false accusation	Victim protection vs. false accusation concerns
Dating and Marriage	low birth rate, singlehood, hypergamy discourse, marriage costs	Demographic anxiety and patriarchy critique
Gender Ideology Conflict	Megalia, reverse discrimination, gender equality debates	Ideological polarization
Beauty Norms	appearance labor, cosmetic surgery, Escape the Corset movement	Gendered appearance expectations
Electoral and Generational Conflict	elections, military compensation, female conscription, fairness discourse	Gendered political identity and fairness discourse

Table 1: Thematic classification of YouTube gender-related discourse domains in South Korea.

The five topical domains reflect major debates in South Korea's contemporary online gender discourse. The Sexual Violence domain emerged from highly politicized debates surrounding hidden-camera crimes and illegal filming, with feminist communities emphasizing victim protection and structural misogyny, while anti-feminist communities often focus on concerns about false accusations against men. The Dating and Marriage domain reflects demographic anxieties linked to South Korea's declining birth rate and debates surrounding the feminist "4B movement" (rejecting marriage, childbirth, dating, and sexual relationships). The Gender Ideology Conflict domain captures direct disputes over gender equality, including accusations of reverse discrimination and competing interpretations of feminism. The Beauty Norms and Grooming domain centers on the feminist "Escape the Corset" movement, which challenges rigid beauty standards, alongside anti-feminist responses that reinforce traditional appearance norms. Finally, the Electoral and Generational Conflict domain reflects the increasing politicization of gender identity in South Korean electoral politics, particularly the growing support for conservative parties among young men and the role of YouTube as a platform for ideological mobilization and political commentary on gender issues.

From this broader corpus, we selected 20 seed videos to initiate recommendation collection. Selection criteria included clear ideological positioning (feminist versus anti-feminist), topical relevance to gender discourse, visibility and engagement metrics, as well as recency of publication. The final seed set consisted of 10 feminist-oriented and 10 anti-feminist-oriented videos. Each domain was represented by two seed videos for each ideological category.

2.2 Sock-Puppet Account Construction

To examine how prior viewing histories shape YouTube's recommendation environment, we constructed three user conditions representing distinct exposure histories. To maintain geographic consistency across all conditions, all browsing sessions were conducted using a VPN configured to a South Korean IP address. The baseline condition approximated a minimal-personalization environment. Recommendations were collected using Google Chrome in incognito mode without any logged-in account. This configuration reduced the influence of accumulated watch history, although some platform-level contextual signals may still have influenced recommendation outputs. The baseline condition therefore functioned as an approximate "cold-start" reference point for comparison across conditions.

To simulate personalized recommendation environments, two persistent YouTube accounts were created using separate logged-in Google accounts. One account was conditioned through exposure to feminist content related to gender equality discourse in South Korea, while the other was conditioned through exposure to South Korean anti-feminist and men's-rights-oriented content. Conditioning material was identified through Korean-language keyword searches and selected based on ideological and topical relevance (see Supplementary Material A for the full training log). Each profile underwent a structured four-day training phase. During each session, lasting approximately 60–90 minutes, we manually logged into the accounts and repeatedly searched for and watched ideologically aligned Korean-language videos.

This process was designed to establish distinct viewing histories and encourage the platform's recommendation system to associate each account with a particular ideological profile.

2.3 Data Collection and Categorization

Recommendation data were collected after the completion of the account-trainning phase. For each seed video, the top 20 videos recommended by YouTube were retrieved under each experimental condition. In the baseline condition, recommendations were collected through a fresh incognito browser for each seed video, whereas conditioned-profile recommendations were collected while logged into the corresponding YouTube accounts in order to preserve accumulated viewing histories (but we put the seed videos with the same ideology as the account in the front in our loop to preserve the consistency of the ideological profile). The dataset contained 1200 connections from seed videos to recommendation videos. Metadata including video ID, title, description, publication date, view count, and like count were extracted for all recommended videos. Because many videos appeared repeatedly across recommendation connections, the final metadata sample consisted of 615 unique recommendation videos.

To examine the characteristics of recommended content, videos were classified according to both ideological categories and topical framing by Large Language Models. Ideological categories included feminism, anti-feminism, neutral, and ambiguous. Topic domains consisted of sexual violence, dating and marriage, gender ideology conflict discourse, political generational and military-related discourse, beauty norms and grooming culture, and other. Content classification was conducted using a human–machine annotation approach. An initial subset of 11 representative videos was manually coded to develop a structured coding guide and establish category boundaries. We then conduct large-scale classification using GPT-4o-mini, which was provided with both the coding scheme, category definitions, and the 11 manually coded reference examples (see Supplementary Material D for full prompt design and codebook). To assess classification reliability, model outputs were compared against human annotations on a randomly sampled validation subset of 20 videos, resulting in a 90% agreement rate. Videos classified as ambiguous, together with cases receiving medium-confidence predictions (70% accuracy), were manually reviewed and reassigned where necessary.

2.4 Ethical Consideration

This study relies exclusively on publicly accessible YouTube content and does not involve interaction with human participants, private user data, or direct engagement with content creators. All recommendation outputs and metadata analyzed in the study were publicly visible through the platform interface at the time of collection. The study employed controlled sock-puppet accounts to simulate distinct viewing histories and observe differences in recommendation outputs. To minimize disruption to the platform ecosystem, the research used a limited number of accounts, avoided large-scale automated scraping, and restricted activity to ordinary viewing and search behaviors. No content was uploaded, and no interactions with other users were conducted.

3 Results

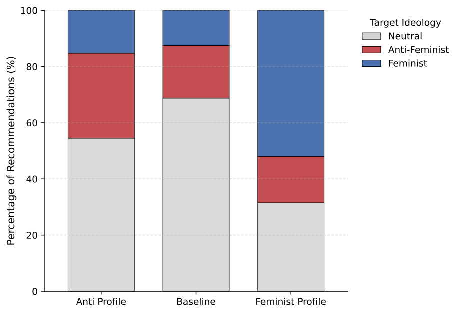
3.1 Ideological Distribution and Directional Bias

Table 2 presents the ideological distribution of YouTube recommendations across the three experimental conditions: the anti-feminist profile, feminist profile, and baseline profile. Overall, the baseline account is characterized by a predominance of neutral recommendations (68.75%), indicating a relatively broad and less ideologically structured recommendation environment prior to any conditioning. In comparison, both trained profiles exhibit a clear shift away from neutrality toward ideologically aligned content.

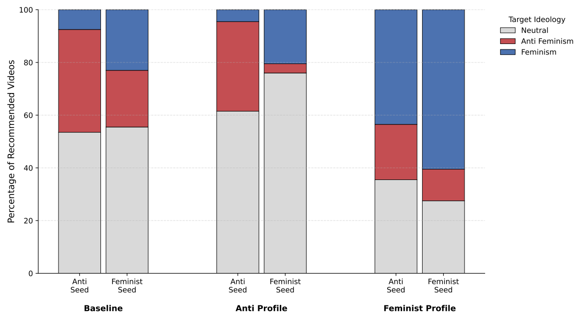
This pattern is further illustrated in Figure 1a. The anti-feminist profile receives a higher proportion of anti-feminist recommendations (30.25%) than the baseline condition (18.75%), while the feminist profile shows a substantially stronger shift toward feminist content (52.00% compared to 12.50% in the baseline). These results indicate that prior viewing histories meaningfully shape following recommendation patterns in ideologically relevant ways.

Condition	Seed Videos	Anti-feminism	Feminism	Neutral	Total
Anti-profile	Total	121 (30.25%)	61 (15.25%)	218 (54.50%)	400
	Anti-feminism	78	15	107	200
	Feminism	43	46	111	200
Baseline	Total	75(18.75%)	50(12.50%)	275 (68.75%)	400
	Anti-feminism	68	9	123	200
	Feminism	7	41	152	200
Feminist profile	Total	66 (16.50%)	208 (52.00%)	126 (31.50%)	400
	Anti-feminism	42	87	71	200
	Feminism	24	121	55	200

Table 2: Ideological distribution of YouTube recommendations across experimental conditions.



(a) Ideological distribution of recommendations.



(b) Ideological transition patterns across conditions.

Fig. 1: Ideological patterns in YouTube recommendations across experimental conditions.

At the same time, the results do not support the existence of fully closed ideological echo chambers. Neutral content remained the largest category within the anti-feminist profile and continued to constitute a large share of recommendations within the feminist profile. In addition, both trained accounts continued to receive some ideologically opposing content. Rather than completely isolating users within a single ideological perspective, the recommendation system appears to reinforce ideological tendencies while still maintaining partial exposure to neutral and cross-cutting material.

Figure 1b reveals clear directional bias in YouTube's recommendation system across different profile conditions. Within the baseline account, anti-feminist seed videos generated more anti-feminist recommendations (34.0%) than feminist seed videos generated feminist recommendations (20.5%). This suggests that anti-feminist content was comparatively more visible within the untrained recommendation environment. However, the feminist profile exhibits the strongest reinforcement, with feminist seeds generating 60.5% feminist recommendations and even anti-feminist seeds still producing a majority of feminist content (43.5%). By contrast, reinforcement in the anti-feminist profile is weaker and more balanced, indicating that feminist conditioning produces more consistent algorithmic alignment than anti-feminist conditioning.

Taken together, these findings suggest that YouTube's recommendation system does not reinforce ideological orientations equally across conditions. Instead, feminist profile conditioning produced a stronger

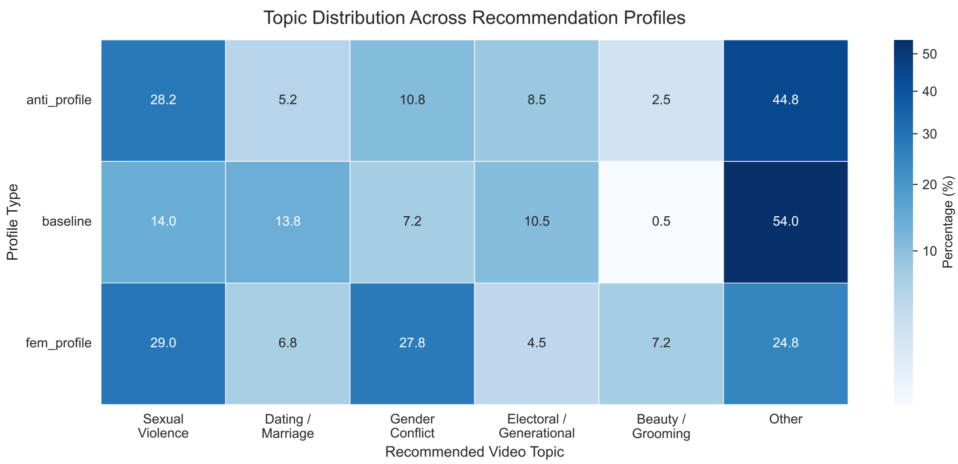


Fig. 2: Topic distribution of YouTube recommendations per condition.

algorithmic reinforcement, while anti-feminist content appeared more visible in the broader baseline recommendation environment.

3.2 Thematic Structure and Topic Reconfiguration

Beyond ideological orientation, Figure 2 also demonstrates that algorithmic personalization reshapes the thematic structure of recommendations. Compared to the baseline profile, both trained conditions show a substantial increase in Sexual Violence content, suggesting that engagement with gender-related material directs the system toward more emotionally and controversially areas. In contrast, the baseline condition remains dominated by heterogeneous “Other” content (54.0%), reflecting a less concentrated thematic environment.

Among the trained profiles, the feminist condition shows the strongest thematic concentration, particularly in Gender Conflict Discourse (27.75%), which is substantially higher than both the anti-feminist profile (10.75%) and the baseline (7.25%). This pattern suggests that feminist conditioning leads to a recommendation environment more strongly structured around explicit ideological confirmation and identity-based gender debates. The feminist profile also shows increased exposure to Beauty Norms and Grooming content, consistent with discourse surrounding the “Escape the Corset” movement. By contrast, the anti-feminist profile remains comparatively diffuse, with a larger proportion of “Other” content, indicating weaker thematic consolidation.

Dating and Marriage content appears to be more prominent in the baseline condition (13.75%) than in either the anti-feminist (5.25%) or feminist (6.75%) profiles, suggesting that in the absence of ideological conditioning this topic remains more stable within the recommendation environment. This reduction is accompanied by evidence of topic reallocation: in the anti-feminist profile, Dating and Marriage content is partially redirected to Sexual Violence, while in the feminist profile it shifts to Gender Conflict Discourse (see Appendix A). This suggests that relatively neutral relational topics become reinterpreted through more ideologically charged frames under conditions of personalization.

However, Electoral and Generational Conflict content were less visible across all conditions. One possible explanation is that explicitly political gender discourse on YouTube may be more tied to specific political events or election cycles, whereas topics such as sexual violence and gender conflict are more continuously embedded within everyday online engagement. In addition, political tensions surrounding gender may frequently appear indirectly through broader discussions of discrimination, demographic anxiety, or interpersonal conflict rather than through explicitly electoral framing. As a result, the recommendation system may more consistently amplify emotionally engaging social conflict content than formal political commentary.

Overall, the results indicate that the recommendation algorithm on YouTube not only shapes ideological alignment but also restructures the topic composition of recommendations. While personalization strengthens exposure to ideologically coherent content, it also concentrates attention on emotionally and socially conflictive topics, rather than producing fully isolated ideological environments.

4 Discussion

4.1 Interpretation

This study contributes to the broader debate on algorithmic personalization by providing empirical insights into how recommendation systems operate in a highly polarized, non-Western context, where identity formation and ideological reinforcement play a central role [21, 26, 27]. Using a sock-puppet experimental research design we compared how YouTube’s recommendation algorithm structures the information environments around feminist and anti-feminist content in South Korea’s gender conflict. Specifically, we examined whether YouTube’s video recommendations for two separately trained profiles—one exposed to South Korean feminist content and the other to anti-feminist content in the same national context—differ in their ideological and topical directional bias.

The findings reveal three key takeaways. First, our results provide only limited empirical support for the existence of echo chambers on YouTube. While recommended content shows strong ideological reinforcement for both the feminist and the anti-feminist profile, the algorithm also introduces a substantial amount of neutral and some ideologically opposing content, thus mitigating the risk of complete echo chambers. This aligns with recent empirical work suggesting that fully isolated echo chambers are rare in platform environments and that exposure diversity might coexist with algorithmic personalization[11, 13, 14].

Second, contrary to our expectations, the results indicate that anti-feminist recommendation environments do not exhibit a stronger filter bubble effect than feminist ones. Instead, the opposite pattern emerged: the feminist profile displayed substantially stronger ideological reinforcement, with recommendations remaining predominantly feminist even when the profile engaged with anti-feminist seed videos. This finding contrasts with Park and Park's [24] study of South Korean news content, which found stronger selective exposure among consumers of conservative media. Overall, our results further complicate the already contradictory evidence about ideological directional bias on YouTube [22–24], and suggest that patterns of algorithmic reinforcement may vary across issue domains and local contexts. Consequently, ideological reinforcement on social media platforms like YouTube cannot be understood solely in terms of a left–right political spectrum, but may depend on a multitude of different characteristics.

Lastly, our LLM assisted topic categorization of recommendation videos reveal several patterns that are worth highlighting. Most strikingly, both the feminist and the anti-feminist profile receive much more content about sexual violence than the neutral baseline profile. Furthermore, the algorithm pushes the conditioned profiles away from content about dating and marriage and towards content about sexual violence and gender conflict. Hence, under conditions of personalization, YouTube's recommendation algorithm seems to push users towards more emotional and controversial topics. This observation stands in partial contrast to the findings by Cakmak and colleagues [29]. They show that within a controversial and politically sensitive topic, YouTube "suggest videos that are less emotionally charged and more balanced in tone" (p. 171). While our study did not focus on a single contentious issue, it suggests that within broader and highly polarized discursive environments such as South Korea's gender conflict, recommendation systems may operate differently. Rather than moderating exposure, personalization appears to increase the salience of topics associated with conflict, victimization, and moral outrage.

4.2 Limitations and Future Research

Several limitations should be acknowledged when interpreting the findings of this study. First, experimental sock-puppet studies cannot directly observe YouTube's recommendation algorithm. Our results only represent observable recommendation environments under controlled conditions rather than the platform's full internal logic. Although a South Korea–based VPN and standardized browsing setups were used, other personalization signals such as device-level information, browser characteristics, and platform-level contextual factors could not be fully controlled. Furthermore, the sock-puppet audit captures recommendation opportunities rather than actual user behavior, as recommendation links reflect algorithmically generated exposure pathways rather than observed selection or engagement. Lastly, while experimental studies aim to contribute to the common body of literature about recommendation algorithms, updates to the algorithm by the platform developer are usually not accounted for [9, 22, 23]. Thus, while the literature regularly disputes itself, uncertainty remains about how much of the observed inconsistencies can be attributed to changes to the algorithm. The findings should therefore be interpreted as a time-specific snapshot of YouTube's recommendation environment rather than a stable or permanent representation of platform behavior.

However, the findings of this study also point toward several promising directions for future research. First, the results suggest that experimental studies of algorithmic curation can benefit from focusing on issue-specific and localized contexts, where political and cultural dynamics may shape recommendation systems in distinct ways. Future work should expand this approach to other contexts, by employing comparable and systematic methods to facilitate cross-case analysis. Second, this study highlights the value of examining not only ideological recommendation patterns but also topical sorting and agenda-setting effects within recommendation systems. More systematic investigations of how algorithms distribute attention across issue domains could provide a deeper understanding of algorithmic influence. Finally, future sock-puppet experiments should employ multiple trained accounts for each ideological condition in order to improve robustness, reduce account-specific effects, and strengthen the validity of findings.

5 Conclusion

Despite growing evidence that fully isolated echo chambers are rare on social media platforms like YouTube, algorithmic personalization can amplify existing polarization, particularly in contexts characterized by strong identity-based conflicts. Using South Korea's gender conflict as a case study, this research examined whether YouTube's recommendation system exhibits ideological and topical directional bias across feminist and anti-feminist recommendation environments. The findings reveal evidence of both. Feminist-conditioned profiles exhibited stronger ideological reinforcement than anti-feminist profiles. In addition, personalization systematically altered the topical composition of recommendations, increasing exposure to content related to sexual violence and gender conflict. These findings suggest that recommendation systems shape not only which viewpoints users encounter, but also which aspects of a social conflict become most salient. More broadly, the study demonstrates the value of moving beyond binary questions of "echo chambers versus no echo chambers" toward a more nuanced understanding of how algorithms structure information environments through both ideological and topical forms of curation.

Response to the reviewers. We received very helpful feedback from two of our fellow students. Both reviewers approved our general research design, and reviewer B further highlights the methodological transparency and relevance of our study. Specifically, they appreciated our regional focus on South Korea, as well as our nuanced analysis of echo chambers using gender ideology and topic salience as indicators. They also provided multiple helpful suggestions for revision.

Reviewer A made a series of smaller suggestions about the structure of our paper, a missing conceptualization of filter bubbles and the visualization of our findings. In our final version we accounted for this critique, restructured our sections and used the Springer Nature LaTeX authoring template to make our visualizations more clear. Furthermore, reviewer A criticised that a detailed explanation of the training process of our Google accounts is missing—we added this in our methodology and expanded our appendices. A's main point of critique concerned the interpretation of our results, they argued that our (preliminary) numbers showed signs of directional bias which we initially refuted. In our final analysis, we made sure that numbers, graphs and interpretations are well aligned and more comprehensible.

Reviewer B raised one central point of critique: they argued that our draft paper lacks a clearly defined academic contribution. Because our research touches on multiple issues and dimensions related to the echo chamber and polarization literature, we agree that our paper was in need of a more precise presentation of our contributions. Although our research design and findings remain multidimensional, we made the aim of our study and the core of our contributions more explicit.

Code and Data Availability. Replication materials, including the analysis code, processed datasets, and supplementary documentation, are available at:
https://github.com/ameliaxiyuchen-eng/CSS_polarization_SouthKorea

Acknowledgements. The authors used generative artificial intelligence tools to assist with programming support, code debugging, and language editing during manuscript preparation. All substantive research decisions, data analysis, interpretation of findings, and final manuscript revisions were conducted and verified by the authors.

Word count: 4173.

Appendix A Topic Transition Table

This table shows the topic transition from seed videos to recommendation videos. Two seed videos were selected for each topic and each ideology. The first 20 recommendation links were collected for each seed video.

Condition	Source Topic	Beauty Norms	Dating Marriage	Electoral Conflict	Gender Conflict	Other	Sexual Violence	Total
anti	beauty	7	5	4	14	32	18	80
anti	dating	2	15	3	11	32	17	80
anti	electoral	1	0	19	8	41	11	80
anti	gender	0	1	5	8	49	17	80
anti	sexual violence	0	0	3	2	25	50	80
baseline	beauty	1	1	5	0	72	1	80
baseline	dating	0	46	1	6	22	5	80
baseline	electoral	1	4	30	9	36	0	80
baseline	gender	0	3	3	13	58	3	80
baseline	sexual violence	0	1	3	1	28	47	80
fem	beauty	13	4	1	19	26	17	80
fem	dating	9	17	1	23	15	15	80
fem	electoral	1	3	13	25	20	18	80
fem	gender	2	1	2	30	25	20	80
fem	sexual violence	4	2	1	14	13	46	80

Table A1: Topic transitions between seed videos and recommended videos across experimental conditions.

References

[1] Pariser, E.: The Filter Bubble: What the Internet Is Hiding from You. Penguin Press, New York (2011)

[2] Sunstein, C.R.: #Republic: Divided Democracy in the Age of Social Media. Princeton University Press, Princeton Oxford (2018)

[3] Cho, J., Ahmed, S., Hilbert, M., Liu, B., Luu, J.: Do Search Algorithms Endanger Democracy? An Experimental Investigation of Algorithm Effects on Political Polarization. Journal of Broadcasting & Electronic Media **64**(2), 150–172 (2020) https://doi.org/10.1080/08838151.2020.1757365

[4] O'Callaghan, D., Greene, D., Conway, M., Carthy, J., Cunningham, P.: Down the (White) Rabbit Hole: The Extreme Right and Online Recommender Systems. Social Science Computer Review **33**(4), 459–478 (2015) https://doi.org/10.1177/0894439314555329

[5] Joe, W., Seán, L., Alastair, R., Fabio, V.: Recommender Systems and the Amplification of Extremist Content. Alexander von Humboldt Institute for Internet and Society gGmbH (2021). https://doi.org/10.14763/2021.2.1565

[6] Yesilada, M., Lewandowsky, S.: Systematic review: YouTube recommendations and problematic content. Internet Policy Review **11**(1) (2022) https://doi.org/10.14763/2022.1.1652

[7] Liu, N., Hu, X.E., Savas, Y., Baum, M.A., Berinsky, A.J., Chaney, A.J.B., Lucas, C., Mariman, R., De Benedictis-Kessner, J., Guess, A.M., Knox, D., Stewart, B.M.: Short-term exposure to filter-bubble recommendation systems has limited polarization effects: Naturalistic experiments on YouTube.

Proceedings of the National Academy of Sciences **122**(8), 2318127122 (2025) https://doi.org/10.1073/pnas.2318127122

[8] Erickson, J.: Rethinking the filter bubble? Developing a research agenda for the protective filter bubble. Big Data & Society **11**(1), 20539517241231276 (2024) https://doi.org/10.1177/20539517241231276

[9] Yu, X., Haroon, M., Menchen-Trevino, E., Wojcieszak, M.: Nudging recommendation algorithms increases news consumption and diversity on YouTube. PNAS Nexus **3**(12), 518 (2024) https://doi.org/10.1093/pnasnexus/pgae518

[10] Nyhan, B., Settle, J., Thorson, E., Wojcieszak, M., Barberá, P., Chen, A.Y., Allcott, H., Brown, T., Crespo-Tenorio, A., Dimmery, D., Freelon, D., Gentzkow, M., González-Bailón, S., Guess, A.M., Kennedy, E., Kim, Y.M., Lazer, D., Malhotra, N., Moehler, D., Pan, J., Thomas, D.R., Tromble, R., Rivera, C.V., Wilkins, A., Xiong, B., De Jonge, C.K., Franco, A., Mason, W., Stroud, N.J., Tucker, J.A.: Like-minded sources on Facebook are prevalent but not polarizing. Nature **620**(7972), 137–144 (2023) https://doi.org/10.1038/s41586-023-06297-w

[11] Hartmann, D., Wang, S.M., Pohlmann, L., Berendt, B.: A systematic review of echo chamber research: Comparative analysis of conceptualizations, operationalizations, and varying outcomes. Journal of Computational Social Science **8**(2), 52 (2025) https://doi.org/10.1007/s42001-025-00381-z

[12] Ledwich, M., Zaitsev, A., Laukemper, A.: Radical bubbles on YouTube? Revisiting algorithmic extremism with personalised recommendations. First Monday (2022) https://doi.org/10.5210/fm.v27i12.12552

[13] Dahlgren, P.M.: A critical review of filter bubbles and a comparison with selective exposure. Nordicom Review **42**(1), 15–33 (2021) https://doi.org/10.2478/nor-2021-0002

[14] Brown, M.A., Bisbee, J., Lai, A., Bonneau, R., Nagler, J., Tucker, J.A.: Echo Chambers, Rabbit Holes, and Algorithmic Bias: How YouTube Recommends Content to Real Users. Social Science Research Network, Rochester, NY (2022). https://doi.org/10.2139/ssrn.4114905

[15] Bruns, A.: Filter Bubble. Alexander von Humboldt Institute for Internet and Society gGmbH (2019). https://doi.org/10.14763/2019.4.1426

[16] Törnberg, P.: How digital media drive affective polarization through partisan sorting. Proceedings of the National Academy of Sciences **119**(42), 2207159119 (2022) https://doi.org/10.1073/pnas.2207159119

[17] Metzler, H., Garcia, D.: Social Drivers and Algorithmic Mechanisms on Digital Media. Perspectives on Psychological Science: A Journal of the Association for Psychological Science **19**(5), 735–748 (2024) https://doi.org/10.1177/17456916231185057

[18] Figà Talamanca, G., Arfini, S.: Through the Newsfeed Glass: Rethinking Filter Bubbles and Echo Chambers. Philosophy & Technology **35**(1), 20 (2022) https://doi.org/10.1007/s13347-021-00494-z

[19] He, S., Fan, Y.: Emotion as a cross-layer mechanism in filter bubbles: A social-psychological perspective. Frontiers in Psychology **16**, 1740709 (2025) https://doi.org/10.3389/fpsyg.2025.1740709

[20] Geschke, D., Lorenz, J., Holtz, P.: The triple-filter bubble: Using agent-based modelling to test a meta-theoretical framework for the emergence of filter bubbles and echo chambers. British Journal of Social Psychology **58**(1), 129–149 (2019) https://doi.org/10.1111/bjso.12286

[21] Ahmmad, M., Shahzad, K., Iqbal, A., Latif, M.: Trap of Social Media Algorithms: A Systematic Review of Research on Filter Bubbles, Echo Chambers, and Their Impact on Youth. Societies **15**(11), 301 (2025) https://doi.org/10.3390/soc15110301

[22] Ibrahim, H., AlDahoul, N., Lee, S., Rahwan, T., Zaki, Y.: YouTube's recommendation algorithm is left-leaning in the United States. PNAS Nexus **2**(8), 264 (2023) https://doi.org/10.1093/pnasnexus/pgad264

[23] Haroon, M., Wojcieszak, M., Chhabra, A., Liu, X., Mohapatra, P., Shafiq, Z.: Auditing YouTube's recommendation system for ideologically congenial, extreme, and problematic recommendations. Proceedings of the National Academy of Sciences **120**(50), 2213020120 (2023) https://doi.org/10.1073/pnas.2213020120

[24] Park, H.W., Park, S.: The filter bubble generated by artificial intelligence algorithms and the network dynamics of collective polarization on YouTube: The case of South Korea. Asian Journal of Communication **34**(2), 195–212 (2024) https://doi.org/10.1080/01292986.2024.2315584

[25] Kim, J.: The Gender War and the Rise of Anti-family Sentiments in South Korea. In: Schoen, R. (ed.) The Demography of Transforming Families, pp. 183–201. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-29666-6_9

[26] Jeong, E., Lee, J.: We take the red pill, we confront the DickTrix: Online feminist activism and the augmentation of gendered realities in South Korea. Feminist Media Studies **18**(4), 705–717 (2018) https://doi.org/10.1080/14680777.2018.1447354

[27] Jung, H.W.: A new variation of modern prejudice: Young Korean men's anti-feminism and male-victim ideology. Frontiers in Psychology **14**, 1230577 (2023) https://doi.org/10.3389/fpsyg.2023.1230577

[28] Lee, Y.-I.: Blaming feminists, claiming pain: Anti-feminist discourse and electoral mobilization by New Men's Solidarity in South Korea. Women's Studies International Forum **112**, 103159 (2025) https://doi.org/10.1016/j.wsif.2025.103159

[29] Cakmak, M.C., Agarwal, N., Oni, R.: The bias beneath: Analyzing drift in YouTube's algorithmic recommendations. Social Network Analysis and Mining **14**(1), 171 (2024) https://doi.org/10.1007/s13278-024-01343-5

[30] Cakmak, M.C., Agarwal, N., Dagtas, S., Poudel, D.: Unveiling Bias in YouTube Shorts: Analyzing Thumbnail Recommendations and Topic Dynamics. In: Thomson, R., Hariharan, A., Renshaw, S., Al-khateeb, S., Burger, A., Park, P., Pyke, A. (eds.) Social, Cultural, and Behavioral Modeling, pp. 205–215. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-72241-7_20

[31] Koo, J.-W., Suh, C.S., Chung, J.W., Sohn, K.-A., Han, K.: The hatred of all against all? Evidence from online community platforms in South Korea. PLoS ONE **19**(5) (2024) https://doi.org/10.1371/journal.pone.0300530